

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Kartheeka Repalle (192372289) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

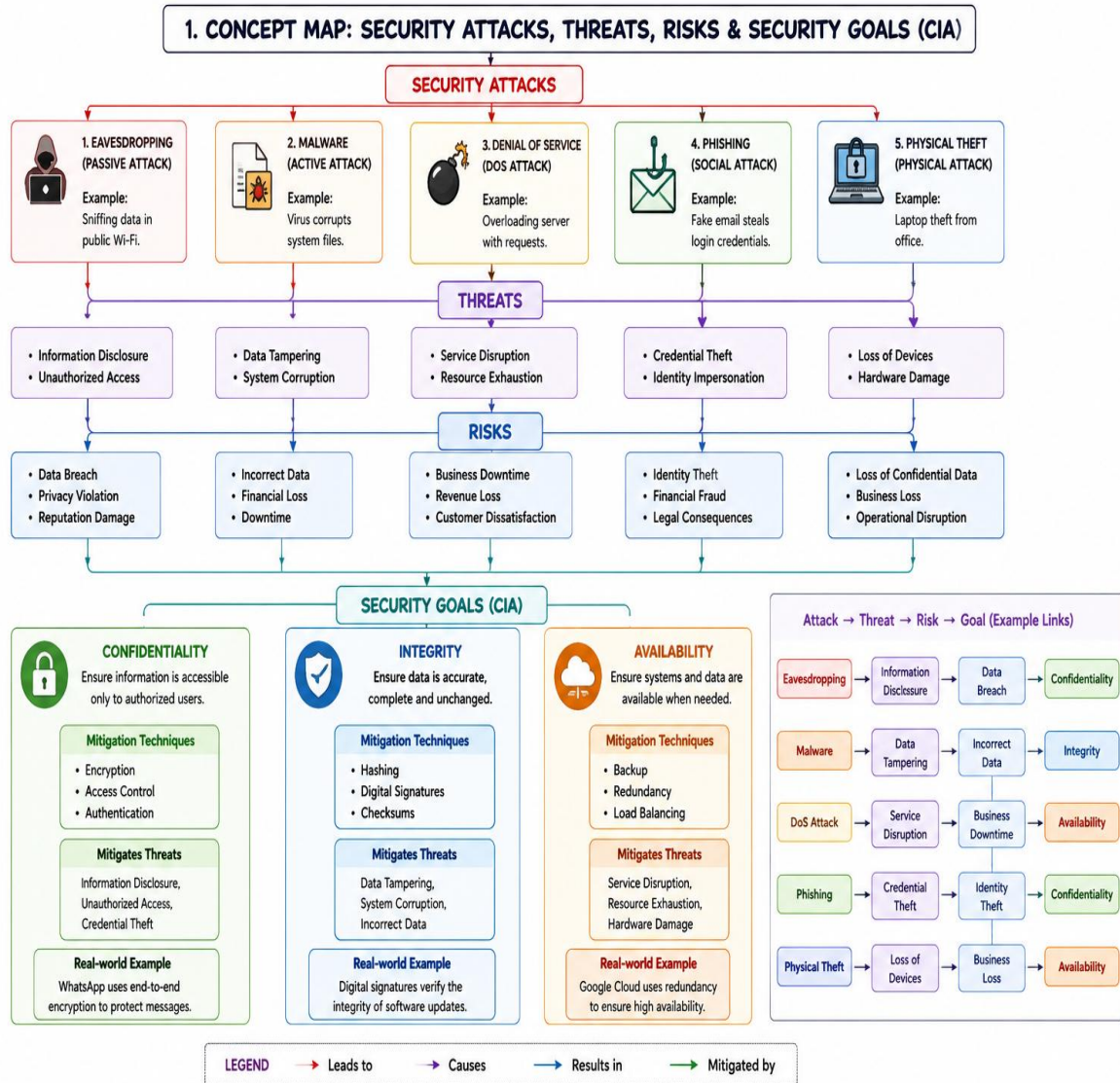
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kartheeka Repalle

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:

- Show how each attack leads to specific threats and risks
- Map how security goals help mitigate them Provide one real-world example for each link



a) Attack → Threat → Risk

Attack	Threat	Risk
Eavesdropping	Information disclosure	Data breach
Malware	File corruption	Financial loss
DoS Attack	Service interruption	Business downtime
Phishing	Credential theft	Identity theft
Physical theft	Device compromise	Loss of confidential data

b) Security Goals and Mitigation

Security Goal	Mitigates	Example
Confidentiality	Unauthorized access	AES encryption protects online banking data
Integrity	Data modification	Digital signatures verify software downloads
Availability	Service interruption	Cloud backups restore services after ransomware

Real-World Examples

Link	Example
Attack → Threat	Phishing email steals login credentials
Threat → Risk	Stolen credentials lead to bank fraud
Confidentiality	WhatsApp end-to-end encryption
Integrity	Git commits verified using digital signatures
Availability	Google Cloud automatic backups

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

Comparison Table

Technique	Key Usage	Complexity	Security
Caesar	Shift value	Very Low	Very Weak
Monoalphabetic	One substitution alphabet	Low	Weak
Vigenère	Keyword	Medium	Good
Hill	Matrix key	High	Strong

Analysis

Caesar Cipher

- Single shift key
- Only 25 possible keys
- Easily broken using brute force

Monoalphabetic Cipher

- One substitution alphabet
- Vulnerable to frequency analysis

Vigenère Cipher

- Uses multiple alphabets
- Harder than Caesar
- Resistant to simple frequency attacks
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Hill Cipher

- Matrix multiplication
- Encrypts blocks
- Difficult to crack without matrix key

Most Secure

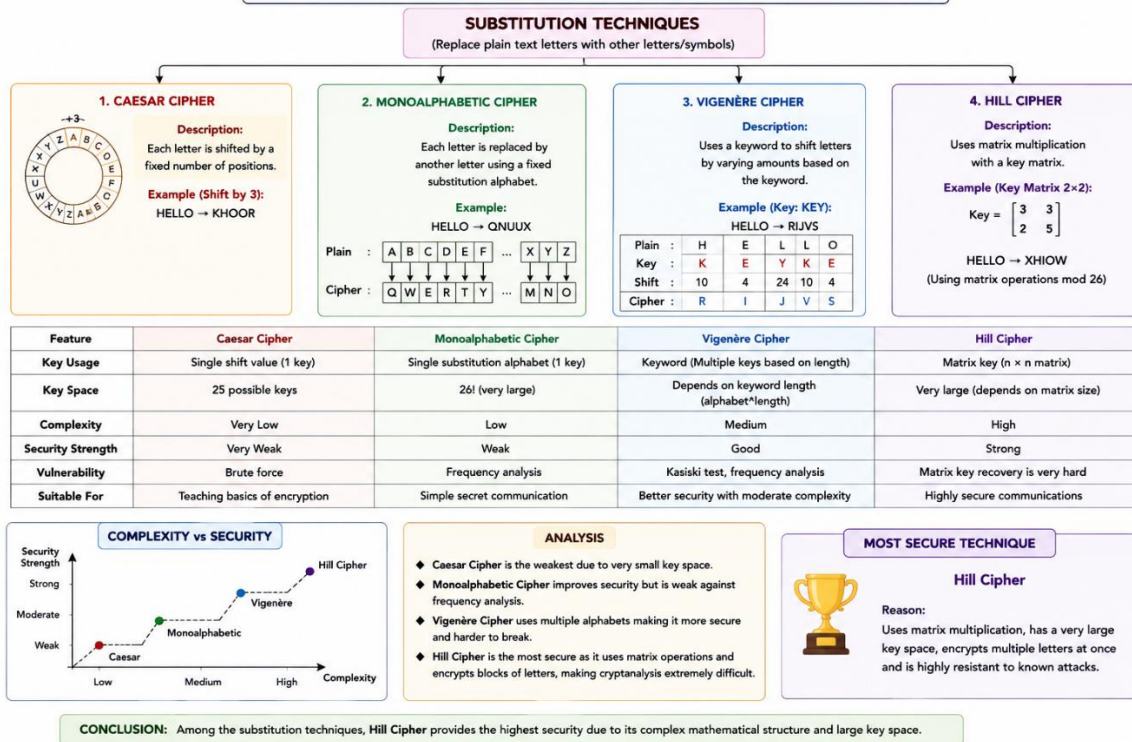
Hill Cipher

Reasons:

- Uses linear algebra
- Matrix-based key
- Encrypts multiple letters simultaneously
- Much larger key space

Annexure A

2. CONCEPT MAP: COMPARISON OF SUBSTITUTION TECHNIQUES



3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

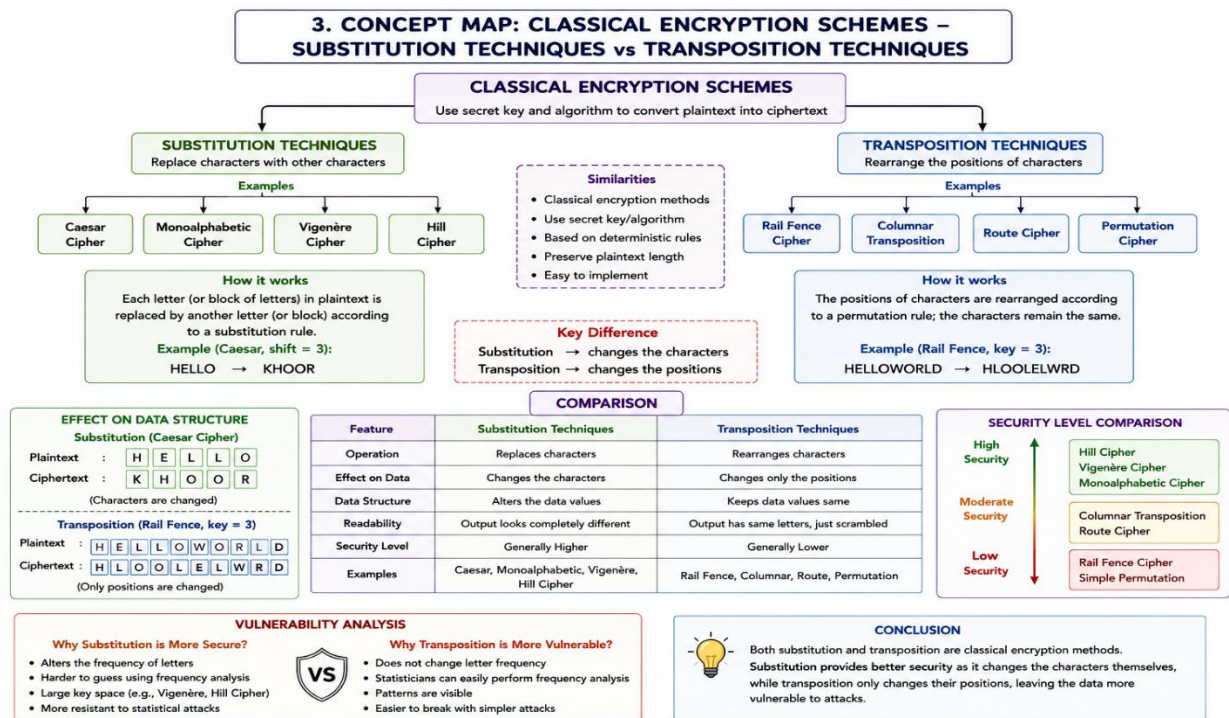
- a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why

Similarities

- Classical encryption methods
- Secret-key algorithms
- Easy implementation
- Preserve plaintext length

Differences

Feature	Substitution	Transposition
Operation	Replace characters	Rearrange positions
Data Structure	Character values change	Character order changes
Readability	Completely changed	Same letters remain
Security	Better	Lower



4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- a. Show relationships between these concepts
- b. Analyze how each contributes to secure communication
- c. Include one real-world application for each

Relationships

Contribution to Secure Communication

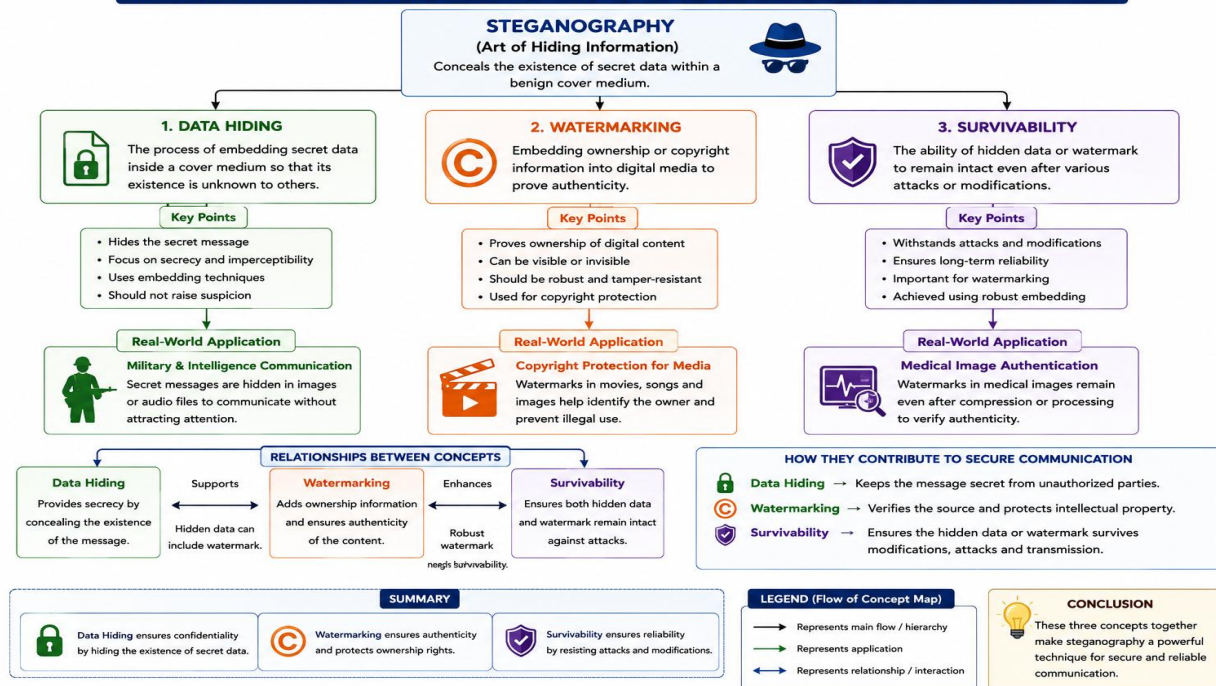
Concept	Contribution
Data Hiding	Prevents detection
Watermarking	Protects ownership
Survivability	Maintains hidden data after modifications

Real-World Applications

Concept	Application
Data Hiding	Secret military communication
Watermarking	Copyright protection for movies
Survivability	Digital medical image authentication

Annexure A

4. CONCEPT MAP: STEGANOGRAPHY – DATA HIDING, WATERMARKING & SURVIVABILITY



5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify

a) Mathematical Connections

Concept	Connection
Extended Euclid	Computes GCD
Modular Inverse	Uses Extended Euclid
Euler Phi	Calculates $\phi(n)$
Euler's Theorem	Uses $\phi(n)$
Fermat's Theorem	Special case of Euler's theorem

b) Role in Encryption

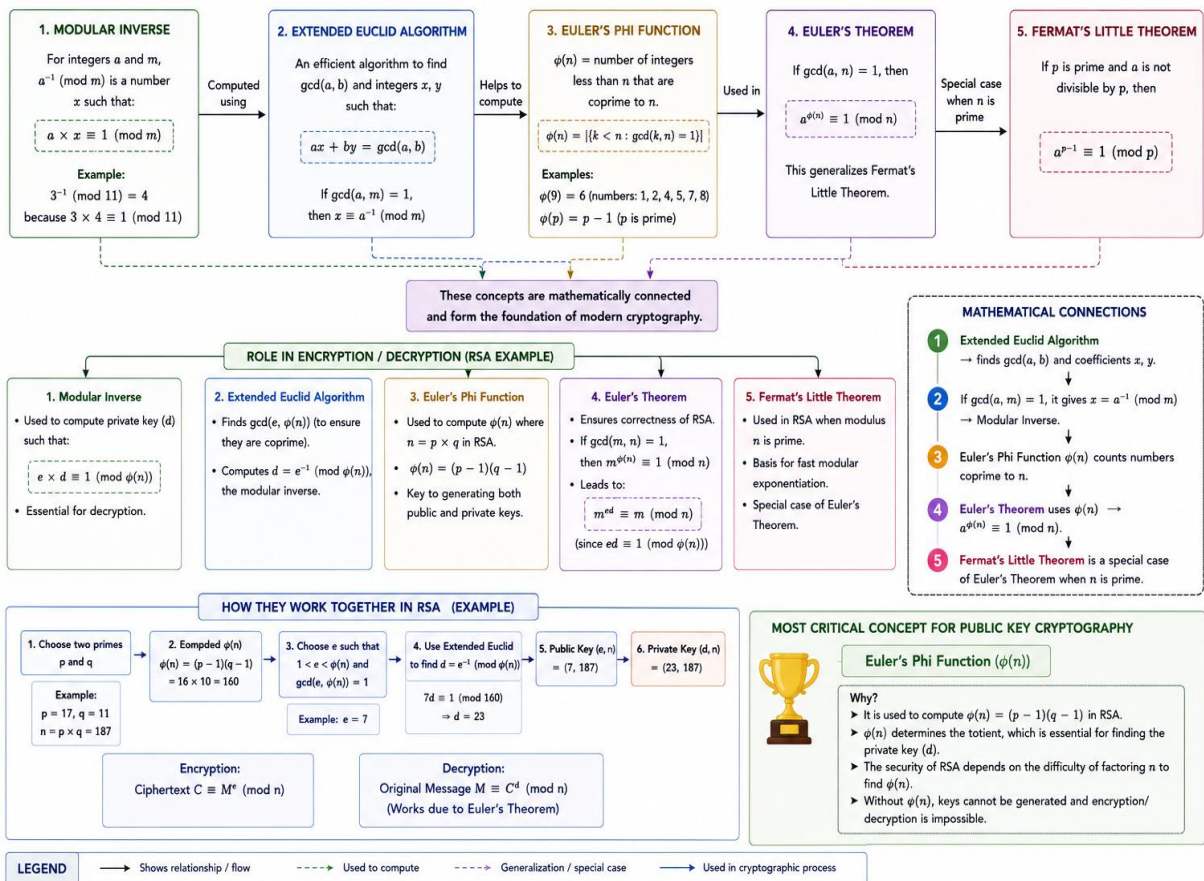
Concept	Role
Extended Euclid	Finds modular inverse for private key
Modular Inverse	RSA decryption key generation
Euler Phi	Computes RSA keys
Fermat	Fast modular exponentiation
Euler's Theorem	Foundation of RSA correctness

c) Most Critical Concept (Euler's Phi Function ($\phi(n)$))

Reasons:

- Used directly in RSA key generation.
- Determines public/private key relationship.
- Needed to compute modular inverse.
- Security depends on difficulty of factoring numbers to calculate $\phi(n)$.

5. CONCEPT MAP: NUMBER THEORY CONCEPTS USED IN CRYPTOGRAPHY



Example

Choose

$$p = 17$$

$$q = 11$$

$$n = p \times q = 187$$

$$\phi(n)$$

$$= (17-1)(11-1)$$

$$= 16 \times 10$$

$$= 160$$

Public key: (e, n)

Private key: $d = e^{-1} \pmod{160}$

The modular inverse is calculated using the **Extended Euclidean Algorithm**, and the correctness of RSA encryption/decryption relies on **Euler's Theorem**.